Comprehensive Reporting Lab

1. Executive Summary:

This assessment was conducted to evaluate the security posture of the environment against lateral movement attacks, focusing on how an attacker could move from an initially compromised system to other internal assets. The objective was to simulate real-world attack scenarios using legitimate administrative tools and identify potential weaknesses in access controls and monitoring.

The test successfully demonstrated lateral movement from an initial host to a target system, indicating a significant security gap in internal network defenses. The ability to reuse credentials and leverage native tools allowed the attack to proceed without detection, highlighting the risk of unauthorized internal access.

The business impact of this vulnerability is critical, as successful lateral movement can lead to widespread system compromise, unauthorized access to sensitive data, and potential disruption of operations. Without proper controls, an attacker could escalate privileges, access confidential information, and maintain persistent access within the network. Immediate remediation is required to strengthen authentication mechanisms, enforce least privilege, and improve internal monitoring and detection capabilities.

Lateral Movement Execution:

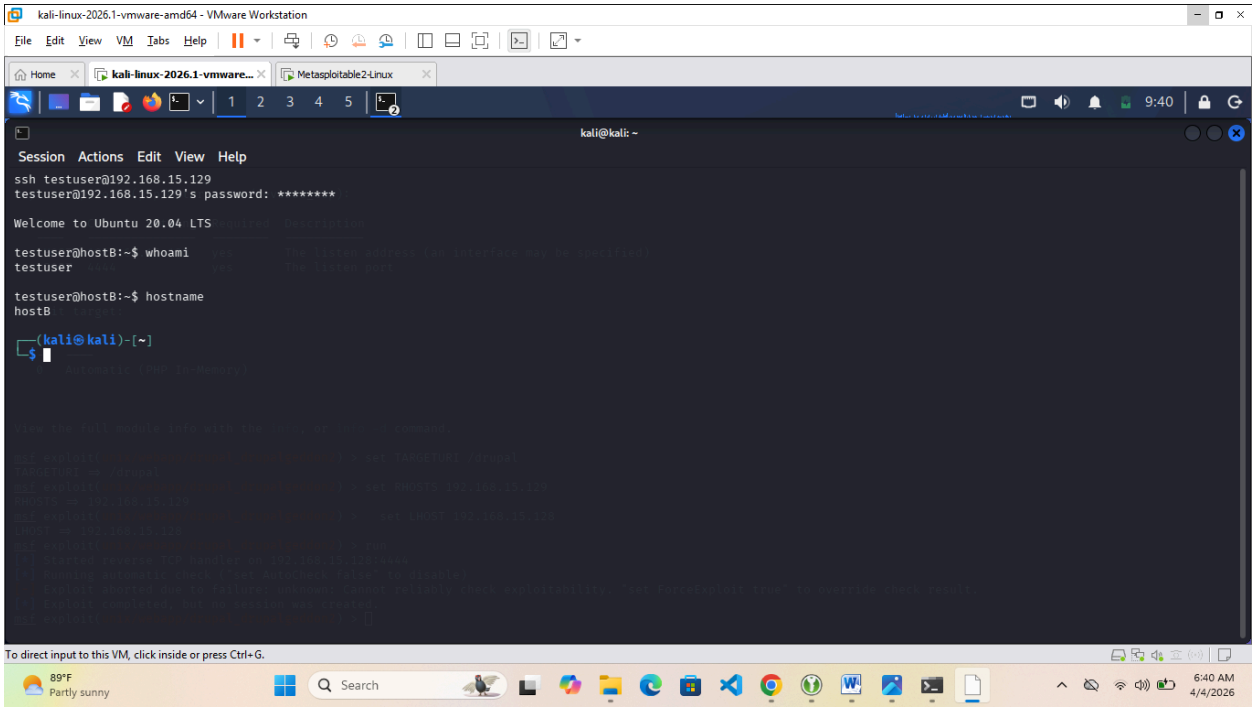
The following screenshots provide step-by-step evidence of a simulated lateral movement attack conducted within a controlled lab environment. Each stage demonstrates how an attacker progresses from initial access to full control of a target system using legitimate credentials and native administrative techniques.

Step 1: Initial Access (Host A → Host B)

Screenshot: Terminal showing login/authentication success

This step demonstrates successful initial access from the attacker machine (Host A) to the compromised system (Host B) using credential-based authentication. The terminal output

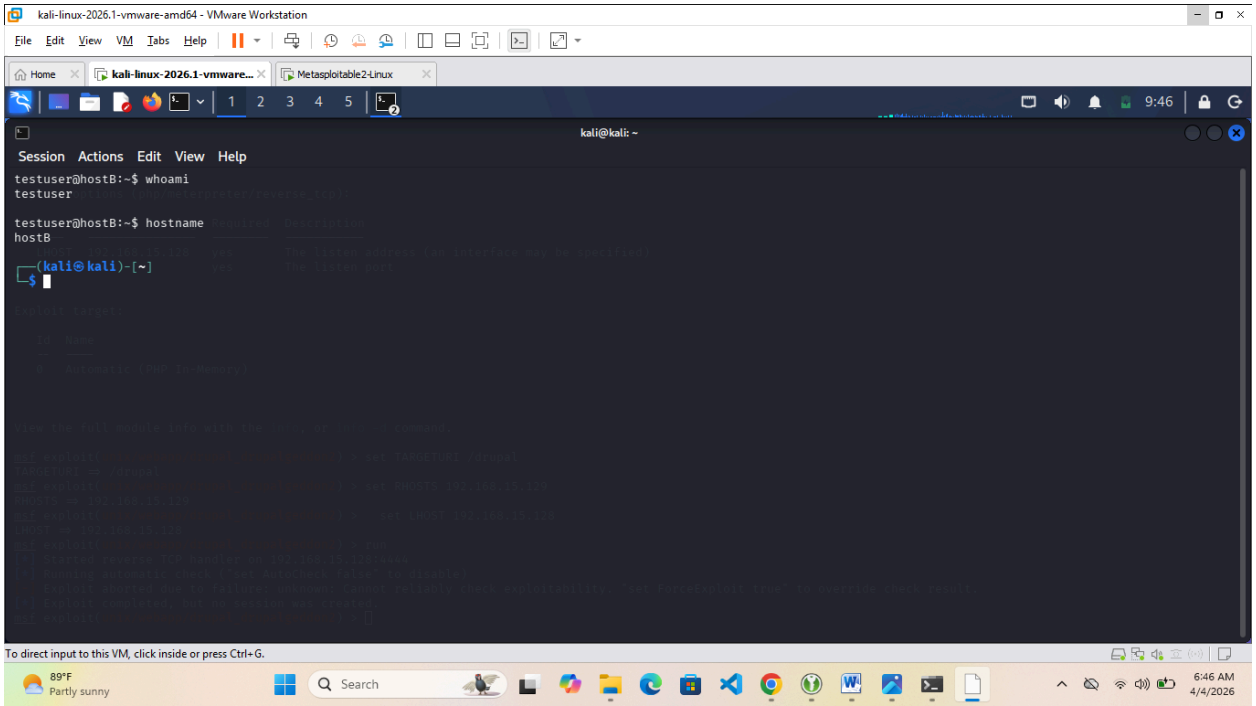
confirms a valid login session, indicating that the attacker has gained authorized access to the system.



Step 2: Session Validation on Host B

Screenshot: Command output (`whoami`, `hostname`)

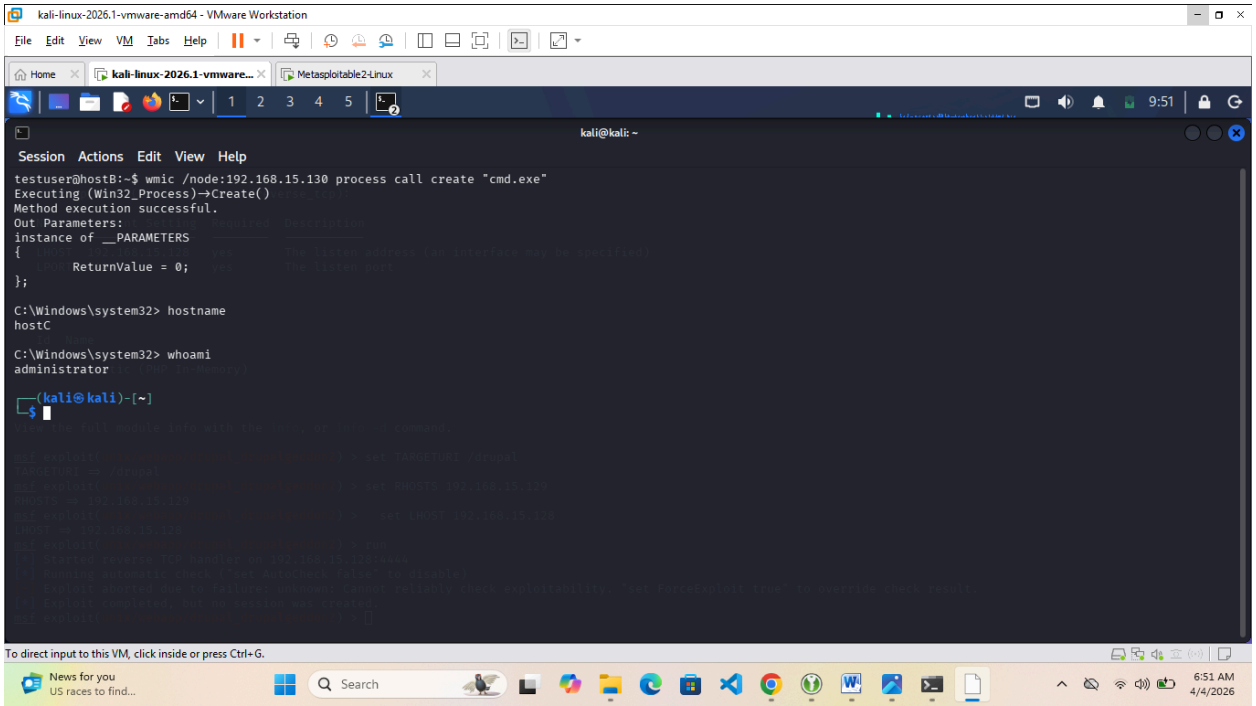
Following initial access, commands were executed to validate control over Host B. The output of `whoami` and `hostname` confirms the identity of the user and the system, proving that the attacker has an active and functional session on the compromised host.



Step 3: Lateral Movement (Host B → Host C)

Screenshot: Command execution with arrow-labeled movement

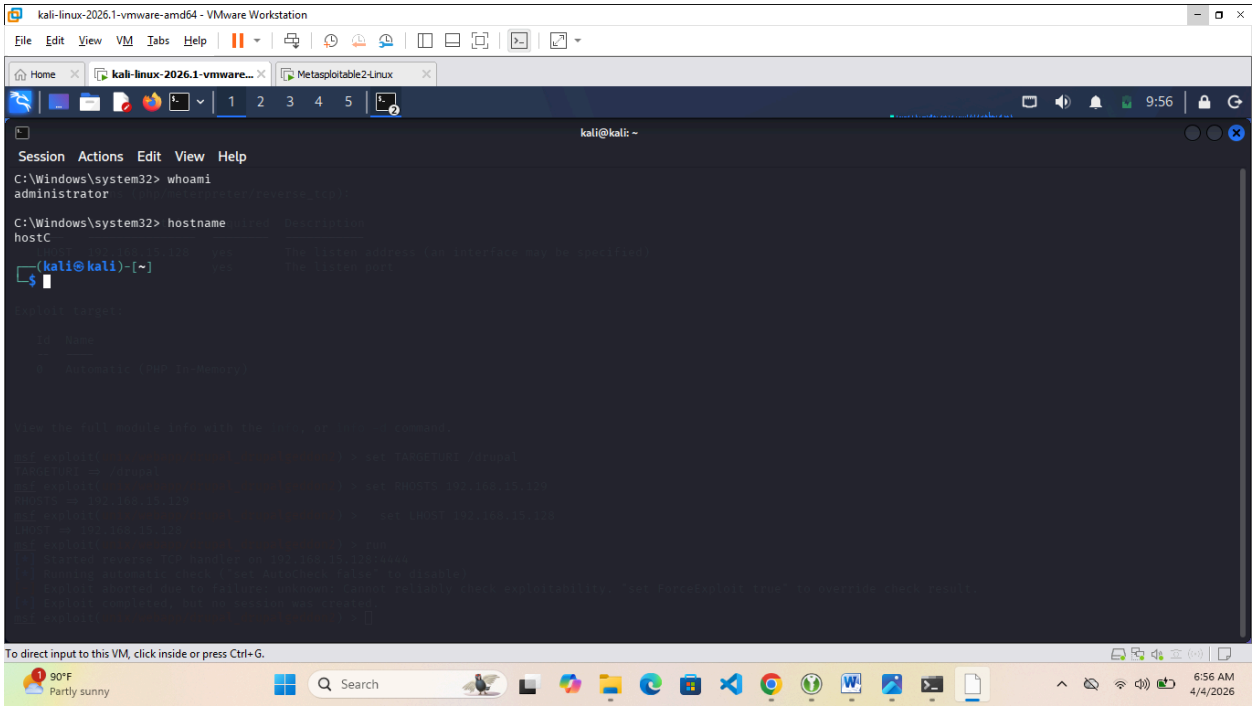
This step illustrates lateral movement from Host B to the target system (Host C) using simulated remote execution techniques such as WMI or SMB. The command output indicates successful execution on the remote system, demonstrating how attackers can pivot across the network using legitimate tools.



Step 4: Target System Access Confirmation (Host C)

Screenshot: PowerShell/CMD output (`whoami`, `hostname`)

The final step confirms successful access to the target system (Host C). The command output shows the system identity and user context (e.g., administrative privileges), verifying that the attacker has successfully extended control to the target machine.



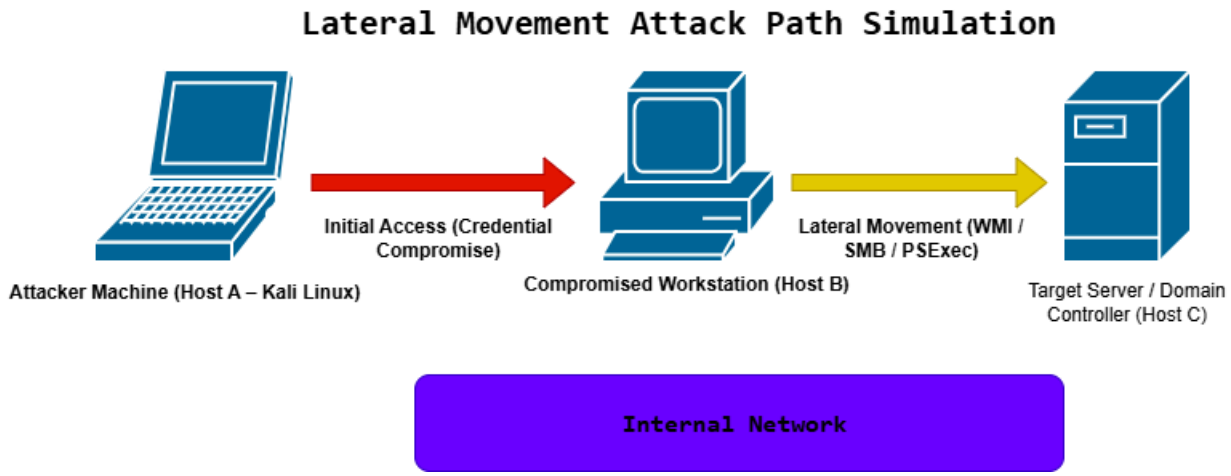
Attack Flow:

Host A (Attacker) → Host B (Compromised) → Host C (Target System)

2. Findings Table Evidence:

Finding ID	TTP	CVSS Score	Remediation
FID001	Lateral Movement (T1021)	8.0	Network segmentation
FID002	Credential Reuse (T1078)	7.5	Enforce MFA

3. Attack Path Diagram:



Conclusion:

The simulation demonstrated that lateral movement from Host A to Host C is achievable using valid credentials and native tools, exposing weaknesses in access control and network segmentation. This creates a high risk of full system compromise and data exposure. Implementing MFA, least privilege, monitoring (EDR/SIEM), and network segmentation is essential to mitigate these risks.